

COMPSCI 1DM3 - Winter 2026

Discrete Mathematics for Computer Science

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Last Lecture:

- Composition of two functions: Given two functions $g : A \rightarrow B$ and $f : B \rightarrow C$, their composition is a function $f \circ g : A \rightarrow C$ with the mapping defined as

$$(f \circ g)(x) = f(g(x))$$

- Floor and Ceiling functions: $\lfloor x \rfloor$, $\lceil x \rceil$
- Properties of floor and ceiling functions:
 - ▶ $\lfloor x \rfloor = n \iff n \leq x < n + 1$
 - ▶ $\lceil x \rceil = n \iff n - 1 < x \leq n$
 - ▶ For any real number $x \in \mathbb{R}$: $x - 1 < \lfloor x \rfloor \leq x \leq \lceil x \rceil < x + 1$
 - ▶ For any integer $n \in \mathbb{Z}$: $\lfloor x + n \rfloor = \lfloor x \rfloor + n$, $\lceil x + n \rceil = \lceil x \rceil + n$
 - ▶ For any real $x \in \mathbb{R}$: $\lfloor -x \rfloor = -\lceil x \rceil$, $\lceil -x \rceil = -\lfloor x \rfloor$

Example

$$\text{Let } n = \lfloor x \rfloor, \boxed{n \leq x < n+1}, n \in \mathbb{Z}$$

Prove that for all $x \in \mathbb{R}$ and $\underline{m \in \mathbb{Z}}$

Case 1: $m > 0$

$$mn \leq \underline{mx} < mn + m$$

$$\lfloor mn \rfloor = mn$$

$$mx < \underline{mn + m}$$

$$\lfloor mx \rfloor \leq mn + m - 1$$

$$mn \leq \lfloor mx \rfloor \leq mn + m - 1$$

$$n \leq \frac{\lfloor mx \rfloor}{m} \leq n + \frac{m-1}{m} < n+1$$

$$\frac{\lfloor mx \rfloor}{m} \in [n, n+1) \rightarrow \left\lfloor \frac{\lfloor mx \rfloor}{m} \right\rfloor = n$$

$$\left\lfloor \frac{\lfloor mx \rfloor}{m} \right\rfloor = \lfloor x \rfloor.$$

Case 2: $m < 0$, let $x = 0.5, m = -1$

$$\lfloor x \rfloor = \lfloor 0.5 \rfloor = 0$$

$$\left\lfloor \frac{\lfloor -1 \times 0.5 \rfloor}{-1} \right\rfloor = \left\lfloor \frac{\lfloor -0.5 \rfloor}{-1} \right\rfloor = \left\lfloor \frac{-1}{-1} \right\rfloor = 1$$

$\therefore$ doesn't hold!

$\therefore x \in \mathbb{R}$ and $m \in \mathbb{Z}^+$

$$\left\lfloor \frac{\lfloor mx \rfloor}{m} \right\rfloor = \lfloor x \rfloor$$

Sec 2.4: Sequences and Sums

- Recall: A set is an unordered collection of distinct objects, e.g., $\{1, 4, 9, 16, 25, \dots, \}$ is the set of all perfect squares.
 - ▶ We can denote this more compactly as follows: $\{a_n\}$, where $a_n = n^2$.

Sec 2.4: Sequences and Sums

- Recall: A set is an unordered collection of distinct objects, e.g., $\{1, 4, 9, 16, 25, \dots\}$ is the set of all perfect squares.
 - ▶ We can denote this more compactly as follows: $\{a_n\}$, where $a_n = n^2$.

$a_1, a_2, a_3, \dots$

Definition

Sequence is the function from a subset of integers (typically $\mathbb{N}$) to a set $\mathcal{S}$

$$\{0, 1, 2, 3, \dots\} \rightarrow \mathcal{S}.$$

Example

Define the function $a : \{0, 1, 2, 3, \dots\} \rightarrow \mathbb{Z}^+$, as $a_n = n^2$.

$$\{a_0, a_1, a_2, a_3, \dots\} = \{0, 1, 4, 9, \dots\}$$

Sequences

- Thus, sequence is the collection of images of $n \in \mathbb{Z}^+$ under a certain function.

Example

Describe the sequence $\{a_n\}$ with $a_n = \frac{1}{n}$ for $n \in \mathbb{Z}^+$.

$$\{a_1, a_2, a_3, \dots\} = \left\{ \frac{1}{1}, \frac{1}{2}, \frac{1}{3}, \dots \right\}$$

Example

Describe the sequence $\{a_n\}$ with $a_n = n + 2$ for $n \in \mathbb{N}$:

$$\{a_0, a_1, a_2, a_3, \dots\} \\ = \{2, 3, 4, 5, \dots\}$$

Sequences

If the domain is a finite set, $\{a_n\}$ is called a finite sequence (or **string**). Otherwise, it is an infinite sequence.

Example

$$\underbrace{\left\{\frac{1}{n}\right\}_{n=1}^{10}} = \{a_1, a_2, \dots, a_{10}\} = \left\{1, \frac{1}{2}, \frac{1}{3}, \dots, \frac{1}{10}\right\}$$

Example

For some $r \in \mathbb{R}$, $\underbrace{\{r^n\}_{n \in \mathbb{N}}}$ $a_n = r^n$

$$\{a_0, a_1, a_2, \dots\} = \{1, r, r^2, r^3, \dots\}$$

Progressions

Geometric progression: Let a and r be real numbers and $r \neq 0$. The sequence $\{a \cdot r^k\}_{k=0}^n$ is the geometric progression:

$$\{a, a \cdot r, a \cdot r^2, a \cdot r^3, \dots, a \cdot r^n\}.$$

Example

The geometric progression for $a = 1$ and $r = -1$ is $\{(-1)^k\}$ or

$$\{1 \cdot (-1)^k\}_{k=0}^n$$

$$\{1, -1, 1, -1, \dots, (-1)^n\}.$$

$$\{(-1)^0, (-1)^1, (-1)^2, (-1)^3, \dots, (-1)^n\}$$

Example

The geometric progression for $a = 2$ and $r = 1/3$ is $\{2(1/3)^k\}$ or

$$\left\{2, \frac{2}{3}, \frac{2}{9}, \frac{2}{27}, \dots, \frac{2}{3^n}\right\}.$$

$$\left\{2\left(\frac{1}{3}\right)^0, 2\left(\frac{1}{3}\right)^1, 2\left(\frac{1}{3}\right)^2, \dots, 2\left(\frac{1}{3}\right)^n\right\}$$

Progressions

Arithmetic progression: Let a and d be real numbers. The sequence $\{a + \textcircled{k}d\}_{k=0}^n$ is the geometric progression:
arithmetic.

$$\{a, a + d, a + 2d, a + 3d, \dots, a + nd\}.$$

Example

The arithmetic progression for $a = -1$ and $d = 4$ is $\{-1 + 4n\}$ or

$$\{-1 + 4k\}_{k=0}^n$$

$$a_n = -1 + 4n$$

Example

The arithmetic progression for $a = 1$ and $d = 3/2$ is $\{1 + 3n/2\}$ or

$$\{1 + \frac{3}{2}k\}_{k=0}^n$$

$$a_n = 1 + \frac{3n}{2}$$

$$\{-1, 3, 7, 11, \dots, -1 + 4n\}.$$

$$\{a_0, a_1, a_2, \dots, a_n\} = \{-1, -1+4, -1+8, \dots, -1+4n\}$$

$$\left\{1, \frac{5}{2}, 4, \frac{11}{2}, \dots, 1 + \frac{3n}{2}\right\}.$$

Recurrence

Sometimes, we can't easily describe all terms in a closed-form expression. Instead, we know a relationship between different terms:

$$\{2, 4, 8, \dots\}$$

Example

$$\underline{a_0 = 2} \text{ and } a_{n+1} = 2a_n$$
$$a_1 = 2a_0 = 2 \cdot 2 = 4$$
$$a_2 = 2a_1 = 2 \cdot 4 = 8$$
$$a_7 = 2a_6$$

Example

$$\underline{a_0 = 1, a_1 = 1} \text{ and } \underline{a_{n+1} = a_n + a_{n-1}} \text{ (the Fibonacci sequence).}$$

$$a_2 = a_1 + a_0 = 2$$

$$a_3 = a_2 + a_1 = 1 + 2 = 3$$

$$a_4 = a_3 + a_2 = 3 + 2 = 5$$

$$\{1, 1, 2, 3, 5, \dots\}$$

Recurrence's solution

Sometimes, we can find a closed-form for a recurrence.

Example

Determine whether $a_n = 3n$ is a solution for the recurrence $a_n = \overbrace{2a_{n-1} - a_{n-2}}^{3n}$.

$$a_{n-1} = 3(n-1) = 3n-3$$

$$a_{n-2} = 3(n-2) = 3n-6$$

$$\begin{aligned} \Rightarrow a_n &= 2(3n-3) - (3n-6) \\ &= 6n-6-3n+6 = 3n \quad \checkmark \end{aligned}$$

Is this solution unique?

No ☺

Recurrence's solution

✓
Show that $a_n = \underline{-n + 2}$ is a solution for the recurrence $a_n = \underline{a_{n-1} + 2a_{n-2} + 2n - 9}$.

$$a_{n-1} = -(n-1) + 2 = -n + 1 + 2 = 3 - n$$

$$a_{n-2} = -(n-2) + 2 = -n + 2 + 2 = 4 - n$$

$$\begin{aligned} \Rightarrow a_n &= (3 - n) + 2(4 - n) + 2n - 9 \\ &= 3 - n + 8 - \cancel{2n} + \cancel{2n} - 9 \\ &= 2 - n \end{aligned}$$

Recurrence's solution

Show that $a_n = -n + 2$ is a solution for the recurrence $a_n = a_{n-1} + 2a_{n-2} + 2n - 9$.

The followings are also solutions:

- $a_n = 5(-1)^n - n + 2$
 - $a_n = 3(-1)^n + 2^n - n + 2$
 - $a_n = 7 \cdot 2^n - n + 2$
- 

For a unique solution, there must be a fixed initial condition.

How to solve a recurrence?

Find a solution for $a_n = 2a_{n-1} - a_{n-2}$ with $\boxed{a_0 = 0 \text{ and } a_1 = 3.}$

$$a_2 = 2a_1 - a_0 = 2 \times 3 - 0 = 6$$

$$a_2 = 3 \times 2 = 6 \quad \checkmark$$

$$a_3 = 2a_2 - a_1 = 2 \times 6 - 3 = 9$$

$$a_4 = 2a_3 - a_2 = 2 \times 9 - 6 = 12$$

$$\{0, \underline{3}, \underline{6}, 9, 12, \dots\}$$

$$3 \times 0 \quad 3 \times 1 \quad 3 \times 2 \quad 3 \times 3 \quad 3 \times 4 \quad \dots$$

$$\underline{a_n = 3n}$$

Summation

Given a sequence $\{a_k\}_{k=0}^n$, we are interested in the sum of all terms, i.e.,

$$S_n = a_0 + a_1 + a_2 + a_3 + \cdots + a_n.$$

Notation:

$$S_n = \sum_{k=0}^n a_k$$

Note that the index of sum is a dummy variable:

$$\sum_{j=0}^n a_j = \sum_{i=0}^n a_i = \sum_{k=0}^n a_k = \sum_{\ell=0}^n a_\ell$$

That is, the sum does not depend on the index.

Summation

Example

What is $\sum_{j=1}^5 j^3$? $= (1)^3 + (2)^3 + (3)^3 + (4)^3 + (5)^3$
 $= 1 + 8 + 27 + 64 + 125 = 225$

Example

What is $\sum_{j=4}^8 (-1)^j$? $= (-1)^4 + (-1)^5 + (-1)^6 + (-1)^7 + (-1)^8$
 $= 1 - 1 + 1 - 1 + 1$
 $= 1$

Change of sum limits

$$\sum_{j=\underline{0}}^{\underline{n}} a_j = \sum_{j=\underline{1}}^{\underline{n+1}} a_{j-1} = \sum_{j=\underline{2}}^{\underline{n+2}} a_{j-2} = \sum_{j=\underline{m}}^{\underline{n+m}} a_{j-m}$$

Geometric progression

Consider the sequence $\{a_k\}_{k=0}^n = \{a \cdot r^k\}_{k=0}^n$, where $a, r \in \mathbb{R}$. Find $\sum_{k=0}^n a_k$.

$$\{a_0, a_1, a_2, \dots, a_n\}$$

$$\{ar^0, ar^1, ar^2, \dots, ar^n\}$$

$$\{a, ar, ar^2, \dots, ar^n\}$$

$$\begin{aligned} \sum_{k=0}^n a_k &= a + ar + ar^2 + \dots + ar^n \\ &= a(1 + r + r^2 + \dots + r^n) \end{aligned}$$